

**FOURTH SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION
APRIL 2022**

Mathematics

ME 4C 04—MATHEMATICAL ECONOMICS

(2014—2018 Admissions)

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes

Total No. of Questions : 20

Maximum : 20 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

ME 4C 04—MATHEMATICAL ECONOMICS

(Multiple Choice Questions for SDE Candidates)

1. r^2 in intercept less model is _____ negative.
(A) Always. (B) Sometimes.
(C) Never. (D) Cannot say.
2. In regression through the origin model, _____ is absent.
(A) The intercept term, β_1 . (B) The slope co-efficient, β_2 .
(C) Error term. (D) Explanatory variables.
3. According to Keynes the value of marginal propensity to consume is :
(A) 0. (B) 1.
(C) 0 and one. (D) Infinity.
4. In the Keynesian linear consumption function $Y = \beta_1 + \beta_2 X$, Y represents :
(A) Income. (B) Consumption expenditure.
(C) Output. (D) Price.
5. In the Keynesian linear consumption function $Y = \beta_1 + \beta_2 X$, the parameters of the model are :
(A) β_1 and β_2 . (B) β_1 and X.
(C) X and Y. (D) Y and β_2 .
6. The term regression was first introduced by :
(A) Irving Fisher. (B) Laspayer.
(C) Francis Galton. (D) Pearson.
7. Regression analysis is concerned with :
(A) Study of the dependence on one variable on the other.
(B) Predicting the average value.
(C) Predicting the population mean.
(D) All of the above.

8. Statistical relationships assumes that variables are :
- (A) Random. (B) Stochastic.
(C) All of the above. (D) None of the above.
9. A model in which regressand is linear and regressor is logarithmic is called ____.
- (A) Regression through the origin. (B) Lin log model.
(C) Log lin model. (D) CLRM.
10. The law of universal regression was first introduced by :
- (A) Irwing Fisher. (B) Laspayer.
(C) Francis Galton. (D) Pearson.
11. If we are studying the dependence of a variable on more than one explanatory variable, the analysis is called :
- (A) Two variable regression analysis.
(B) Multiple regression analysis.
(C) Single regression analysis.
(D) None of the above.
12. The set of all possible outcomes of an experiment or measurement is known as :
- (A) Population. (B) Census.
(C) Sample. (D) Variable.
13. The locus of points conditional means of the dependent variable for the fixed values of the explanatory variables is :
- (A) Venn Diagram. (B) Lorenz curve.
(C) Probability curve. (D) Population regression curve.
14. In the regression function $E(Y/X_i) = \beta_1 + \beta_2 X_i$, regression co-efficients are :
- (A) Y and X. (B) Y and β_1 .
(C) β_1 and β_2 . (D) β_1 and X_i .

15. Which is the assumption of Gaussian standard classical linear regression model ?
- (A) Linear regression model.
 - (B) X values are fixed.
 - (C) Zero mean values for disturbances.
 - (D) All of the above.
16. Homoscedasticity means _____ for disturbances.
- (A) Equal mean.
 - (B) Equal variance.
 - (C) Zero mean.
 - (D) None of the above.
17. In the function, $Q = \alpha + \beta P$, the intercept co-efficient is :
- (A) α .
 - (B) β .
 - (C) P.
 - (D) Q.
18. In the regression context, the OLS estimators are BLUE according to :
- (A) Central Limit Theorem.
 - (B) Gauss Markov Theorem.
 - (C) Young Theorem.
 - (D) Fisher's Theorem.
19. The numerical value of coefficient of correlation lies between :
- (A) -1 and 1.
 - (B) 0 and 1.
 - (C) $-\infty$ to $+\infty$.
 - (D) $-\infty$ to 1.
20. Which of the following is used to measure the degree of association between two variables :
- (A) Co-efficient of determination.
 - (B) Co-efficient of correlation.
 - (C) Standard error.
 - (D) Standard deviation.

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Mathematics

ME 4C 04—MATHEMATICAL ECONOMICIS

(2014—2018 Admissions)

Time : Three Hours

Maximum : 80 Marks

Part A

Answer all the twelve questions.

Each question carries 1 marks.

1. r^2 in intercept less model is _____ negative.
A) Always. B) Sometimes.
C) Never. D) Cannot say.
2. The first step in traditional econometric methodology is :
A) Statement of theory. B) Forecasting.
C) Obtaining data. D) Estimation of the model.
3. In the Keynesian linear consumption function $Y = \beta_1 + \beta_2 X$ Y represents :
A) Income. B) Consumption expenditure.
C) Output. D) Price.
4. Model in which regressand is logarithmic is called _____.
A) Regression through the origin. B) lin log model.
C) Log lin model. D) CLRM :
5. Stochastic variables are those having :
A) Probability distribution. B) Indexation.
C) Correlation. D) Causation.

Turn over

- (12 × 1 = 12 marks)

Answer any six questions in two or three sentences.
Each question carries 3 marks.

13. Write a short note on 'Two Variable Regression Analysis'.

14. What is time series data ?
15. Write three statistical properties of OLS estimators.
16. Show that the sample correlation co-efficient lies between -1 and 1 .
17. If X_1, X_2 and X_3 are uncorrelated variables each having the same standard deviation, show that the co-efficient of correlation between $X_1 + X_2$ and $X_2 + X_3$ is equal to $1/2$.
18. Write any *three* properties of OLS estimators under the normality assumption.
19. What do you mean by a confidence interval ?
20. Given the sample regression function $Y_i = \hat{\beta}_2 X_i + \hat{u}_i$. Find $\hat{\beta}_2$.
21. Consider the regression model $y_i = \beta_1 + \beta_2 x_i + u_i$ where $y_i = (Y_i - \bar{Y})$ and $x_i = (X_i - \bar{X})$. In this case, show that the regression line must pass through the origin.

(6 × 3 = 18 marks)

Part C

Answer any six questions from the following.

Each question carries 5 marks.

22. Discuss briefly about the four measurement scales of variables in the regression analysis.
23. Write a note on the Concept of Population Regression Function (PRF).
24. Consider the sample regression $Y_i = \hat{\beta}_1 + \hat{\beta}_2 X_i + \hat{u}_i$ with the assumptions (i) $\sum \hat{u}_i = 0$; and (ii) $\sum \hat{u}_i X_i = 0$. Obtain the estimators $\hat{\beta}_1$ and $\hat{\beta}_2$.
25. Consider the following formulations of the two-variable PRF : Model I $Y_i = \beta_1 + \beta_2 X_i + u_i$ Model II $Y_i = \alpha_1 + \alpha_2 (X_i - \bar{X}) + u_i$ Find the estimators β_2 and α_2 . Are they identical ? Are their variances identical?
26. Find the Confidence Intervals for Regression Co-efficients β_1 .
27. Write a note on Testing of Hypothesis.

Turn over

28. Find $\hat{\beta}_2$ and $\hat{\beta}_1$ from the following data :

X	1	4	5	6
Y	4	5	7	12

29. Consider the loglinear model : $\ln Y_i = \beta_1 + \beta_2 \ln X_i + u_i$ Plot Y on the vertical axis and X on the horizontal axis. Draw the curves showing the relationship between Y and X when $\beta_2 = 1$, and when $\beta_2 > 1$, and when $\beta_2 < 1$.
30. What is Log Linear regression model ? How to measure elasticity using this model.

(6 × 5 = 30 marks)

Part D

*Answer any two questions from the following.
Each question carries 10 marks.*

31. Discuss the significance of the Stochastic Disturbance Term
32. Discuss the assumptions made in the classical linear regression model.
33. Calculate the correlation co-efficient for the following heights (in inches) of fathers(X) and their sons(Y).

X	65	66	67	67	68	69	70	72
Y	67	68	65	68	72	72	69	71

34. Discuss LogLin and LinLog Models.

(2 × 10 = 20 marks)